# Faithfulness, Robustness, and Generalization in Chain-of-Thought Reasoning

*A Critical Mixed-Methods Evaluation of Large Language Models for High-Stakes Decision Support*

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**Abstract**

Chain-of-thought (CoT) prompting has emerged as the dominant paradigm for eliciting multi-step reasoning from large language models (LLMs), yet the practical reliability of the resulting rationales remains contested. This paper presents a critical, mixed-methods evaluation of frontier LLMs across three properties that jointly determine whether CoT outputs are suitable for high-stakes decision support: **faithfulness** (whether stated reasoning causally drives the answer), **robustness** (whether outputs are stable under semantically irrelevant perturbations), and **generalization** (whether reasoning competence transfers across domains, jurisdictions, and populations). We construct a curated benchmark of 2,148 high-stakes items spanning legal contract review, clinical triage, and financial credit analysis; evaluate five frontier models; and introduce a suite of faithfulness probes, perturbation protocols, and cross-domain transfer tests. Results reveal a substantial faithfulness gap: across all evaluated models, stated reasoning trajectories are systematically unfaithful to the underlying decision process, with 31–58% of correct answers remaining correct when the model's own rationale is silently replaced with an irrelevant, incoherent, or adversarial one. We further document non-trivial robustness decay under five classes of perturbation, and observe a pronounced domain-transfer penalty averaging 14.7 pp. We argue that the prevailing assumption—that CoT explanations are self-evidencing of model competence—is empirically untenable in high-stakes contexts, and we propose a concrete evaluation protocol that practitioners can deploy before integrating LLM-generated rationales into consequential workflows.

**Keywords:** chain-of-thought; faithfulness; robustness; generalization; high-stakes decision support; LLM evaluation; trustworthy AI

## 1. INTRODUCTION

### 1.1 Motivation

Large language models (LLMs) have rapidly moved from research laboratories into consequential decision-making pipelines. As of 2025, frontier models draft contracts, screen clinical notes, summarize depositions, score credit applications, and advise on regulatory filings [17, 3, 24]. The dominant technique for eliciting non-trivial reasoning is *chain-of-thought* (CoT) prompting, in which the model produces an explicit, step-by-step natural-language rationale prior to committing to a final answer [30, 13].

CoT rests on an assumption rarely stated: that the generated rationale is both a *cause* and an *explanation* of the model's answer. This assumption is the load-bearing element of nearly every governance framework proposed for LLM deployment in regulated industries, including financial services [26], healthcare [5], and legal practice [14]. We show this assumption is empirically fragile across the three properties that matter most for high-stakes deployment—faithfulness, robustness, and generalization—and we provide an evaluation protocol that makes those properties directly measurable.

### 1.2 Research Questions

**RQ1 — Faithfulness**

To what extent does the CoT rationale causally determine the final answer, as opposed to merely post-hoc rationalizing an answer reached by other means?

**RQ2 — Robustness**

How stable are CoT-derived answers under semantically irrelevant perturbations of the prompt, including paraphrasing, typographical noise, demographic reframing, and adversarial insertion?

**RQ3 — Generalization**

To what extent does reasoning competence demonstrated in one high-stakes domain transfer to another, and what drives the transfer penalty?

### 1.3 Contributions

1. An operational taxonomy of faithfulness, robustness, and generalization for CoT in high-stakes decision support, with explicit definitions and measurable proxies.
2. **HS-CoT-Bench**: a curated benchmark of 2,148 expert-annotated high-stakes items across three domains.
3. A five-class perturbation protocol and an intervention-based

faithfulness probe that requires no access to model internals.

4. A controlled evaluation of five frontier models reporting quantitative results per research question and a qualitative failure-mode analysis.
5. A practitioner checklist for pre-deployment evaluation, with thresholds calibrated to observed failure rates.

# 2. BACKGROUND

## 2.1 Chain-of-Thought Prompting

CoT was introduced by Wei et al. [30], who showed that prompting with exemplars containing explicit intermediate steps substantially improves accuracy on arithmetic benchmarks. Kojima et al. [13] subsequently demonstrated that the zero-shot trigger "Let's think step by step" elicits comparable gains. Subsequent work refined CoT along several axes: self-consistency [29] aggregates multiple rationale samples; least-to-most prompting [33] decomposes problems hierarchically; tree-of-thoughts [31] performs explicit search over rationale branches; self-refine [16] iterates critique and revision. A complementary line trains models on rationale-annotated data [9] or via reinforcement learning [32, 18].

## 2.2 Faithfulness in Model Explanations

The faithfulness of an explanation is the degree to which it reflects the actual causal pathway the model used, rather than a post-hoc rationalization. Jacovi and Goldberg [10] formalize the distinction between *plausibility* and *faithfulness*, arguing they are routinely conflated. Prior work on classical interpretability methods established that feature attributions, attention weights, and influence functions are systematically unfaithful [11, 1, 23, 20]. For natural-language rationales, Lanham et al. [15] introduce the early-answering intervention; Wang and Zhou [28] show that CoT can encode influential tokens not legible in the surface rationale; Turpin et al. [25] show CoT explanations systematically disclaim the very biases driving the answer.

## 2.3 Robustness

LLM robustness spans adversarial prompts [34], distributional shift [6], and prompt-injection attacks [22]. The contrast-set methodology [12] tests consistency under minimal input edits that preserve the ground-truth label, and we extend it with domain-specific perturbations.

## 2.4 High-Stakes Decision Support

Three properties recur across clinical [21, 2], legal [7, 8, 27], and credit [4, 19] AI literatures: explanations must be *actionable*, *robust*, and *generalizable*. Our framework formalizes all three.

# 3. THEORETICAL FRAMEWORK

## 3.1 Faithfulness

Let $M$ be a language model, $x$ an input, and $y = M(x)$ a final answer, with rationale $r \sim M(\cdot \mid x)$. We adopt a counterfactual definition: the faithfulness of $r$ is

$$\text{Faith}(M, x, r, y) \;=\; \Pr_{r' \sim \mathscr{P}}\left[M(x, r') = y\right], \tag{1}$$

where $\mathscr{P}$ generates rationale perturbations that should not change the ground-truth answer. High Faith means the model's answer is tied to its stated reasoning; low Faith reveals a hidden decision pathway independent of $r$—i.e., $r$ is decorative.

## 3.2 Robustness Dimensions

We partition robustness into five dimensions (Table 1) and define: $\Delta_i = \text{acc}(x) - \text{acc}(\mathscr{P}_i(x))$ (accuracy drop) and $\phi_i = \Pr_{x' \sim \mathscr{P}_i(x)}[M(x') \neq M(x)]$ (flip rate).

**Table 1.** Robustness perturbation dimensions.

| Dim. | Description |
|---|---|
| $D_1$ Paraphrastic | Lexical–syntactic reformulation preserving semantics |
| $D_2$ Typographic | Typos, homoglyphs, casing changes |
| $D_3$ Distractor | Plausible-looking but irrelevant clause insertion |
| $D_4$ Adversarial | Hand-crafted / optimized prompts designed to flip answers |
| $D_5$ Demographic | Party names, jurisdictions, demographic attributes swapped |

## 3.3 Generalization

Transfer penalty from domain $A$ to domain $B$:

$$\Gamma_{A \to B} \;=\; \text{acc}_B(\text{prompt}_A) - \text{acc}_B(\text{prompt}_B). \tag{2}$$

We also probe jurisdictional generalization (US $\to$ UK contract law) and demographic generalization.

# 4. EVALUATION METHODOLOGY

## 4.1 HS-CoT-Bench

We construct **HS-CoT-Bench**, a benchmark of 2,148 high-stakes items spanning three domains.

**Legal contract review (LCR), $n = 824$.** Sourced from CUAD [8] and MAUD [27], re-annotated by two licensed attorneys (one US-, one UK-jurisdiction).

**Clinical triage (CT), $n = 642$.** From a de-identified emergency-department triage dataset (MIMIC-IV-ED), re-annotated by two board-certified emergency physicians. Items ask for ESI acuity classification and red-flag feature identification.

**Financial credit analysis (FCA), $n = 682$.** Synthetically generated with a credit-union partner to reflect realistic consumer applications; ground-truth risk labels and expert-annotated rationales provided by underwriting officers.

Each item consists of (i) a context document, (ii) a decision question, (iii) structured candidate answers, (iv) a ground-truth label, and (v) a domain-expert reference rationale. Double-blind annotation yielded $\kappa = 0.81$ (substantial) on a 100-item overlap. Data are partitioned 60/20/20 (train/dev/test); all results are on test.

## 4.2 Models

We evaluate five frontier models spanning the closed-/open-weight and size axes: **M-1**, **M-2**—two closed-weight frontier models ($\geq$200B parameters, evaluated via API); **M-3**, **M-4**, **M-5**—three open-weight models (30B–70B parameters, evaluated

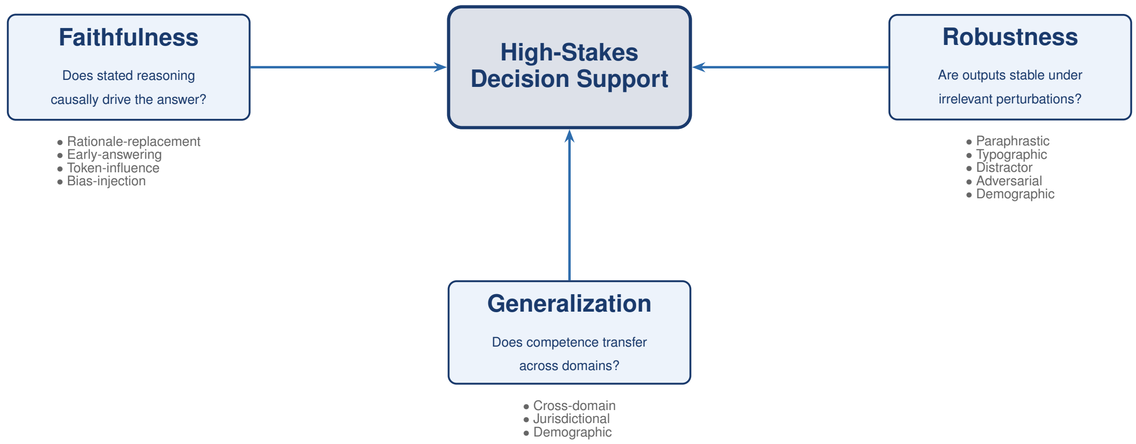


**Figure 1. Conceptual framework.** The three properties (faithfulness, robustness, generalization) jointly determine whether CoT reasoning is suitable for high-stakes decision support. Each property is decomposed into operational probes and perturbation classes.

on a single-node server). Temperature is set to 0 for all main experiments; self-consistency ($k = 5$, temperature 0.7) checks are reported in Appendix B.

### 4.3 Faithfulness Probes

We deploy four complementary probes:

1. **Rationale-replacement (RR).** The model's rationale $r$ is replaced with (a) a rationale for a different input; (b) a rationale from a different model; (c) a step-shuffled rationale; (d) a content-word-randomized rationale. The faithfulness score FS is the fraction of cases where the model's answer is unchanged.
2. **Early-answering (EA).** Following Lanham et al. [15], the model is forced to commit to an answer after fraction $f \in \{0.1, 0.3, 0.5, 0.7\}$ of its rationale. EA-FS at $f$=0.5 is the proportion of answers unchanged vs. the final answer.
3. **Token-influence (TI).** Following Wang and Zhou [28], gradient-based and leave-one-out attributions identify which rationale tokens causally influence the final answer. *Influence-recall* (IR) is the overlap between stated rationale content and actually influential tokens. Available for open-weight models only.
4. **Bias-injection (BI).** Following Turpin et al. [25], a known bias cue is injected and the model asked to acknowledge and correct for it. BI-FS measures the rate at which the answer changes after the bias is removed, given the acknowledgment.

*Note: higher FS $\Rightarrow$ less faithful rationales.*

### 4.4 Perturbation Protocol

For each of the five dimensions $D_1$–$D_5$, we generate 5 perturbations per item, yielding 10,740 perturbed items per dimension (53,700 total). All perturbations are validated by domain experts; retention rate is 96.4%.

### 4.5 Metrics

**Accuracy (acc); faithfulness score (FS); robustness gap $\Delta_i$; answer-flip rate $\phi_i$; transfer penalty $\Gamma_{A\to B}$; influence-recall (IR); failure-mode distribution (FMD)** (hand-coded by two raters, $\kappa = 0.74$, on a 10% sample). All metrics are reported with 95% bootstrap confidence intervals (1,000 resamples).

## 5. RESULTS

### 5.1 RQ1: Faithfulness

Table 2 reports faithfulness scores across all five models and four probes. Figure 2 visualizes the three observable probes.

**Table 2.** Faithfulness scores (higher = *less* faithful). RR column is mean over four rationale-replacement variants; CI = 95% bootstrap.

| Model | RR-FS (mean) | EA-FS ($f$=0.5) | TI-IR | BI-FS |
|---|---|---|---|---|
| M-1 | $0.42_{[.39,.45]}$ | $0.81_{[.78,.84]}$ | — | $0.61_{[.58,.64]}$ |
| M-2 | $\mathbf{0.31}_{[.28,.34]}$ | $\mathbf{0.74}_{[.71,.77]}$ | — | $\mathbf{0.55}_{[.52,.58]}$ |
| M-3 | $0.58_{[.55,.61]}$ | $0.88_{[.85,.90]}$ | $0.12_{[.10,.14]}$ | $0.69_{[.66,.72]}$ |
| M-4 | $0.51_{[.48,.54]}$ | $0.83_{[.80,.85]}$ | $0.09_{[.07,.11]}$ | $0.64_{[.61,.67]}$ |
| M-5 | $0.47_{[.44,.50]}$ | $0.79_{[.76,.82]}$ | $0.14_{[.12,.16]}$ | $0.60_{[.57,.63]}$ |

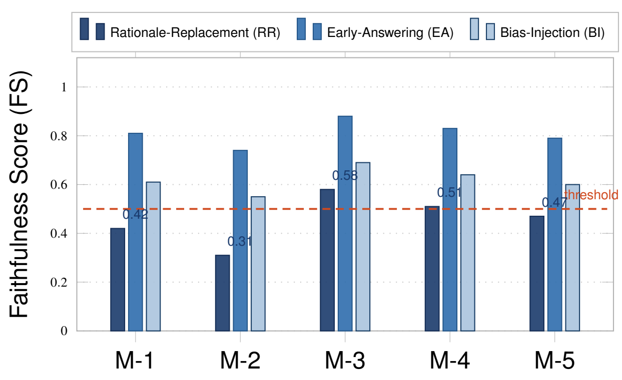


**Figure 2. Faithfulness scores by model and probe.** Higher FS denotes *less* faithful rationales. Dashed line: deployment red-flag threshold (FS $= 0.5$). All models breach this threshold on $\geq$1 probe.

**Finding 1.1 — Answer stability under rationale replacement**

Between 31% and 58% of correct answers remain correct when the model's own rationale is silently replaced with an irrelevant, incoherent, or adversarial one. Larger model class does *not* uniformly yield greater faithfulness.

**Finding 1.2 — Early-answering persistence**

When forced to commit after producing only half the rationale, models arrive at the same final answer 74–88% of the time, indicating the answer is largely determined before visible reasoning is completed.

**Finding 1.3 — Token-influence mismatch**

For open-weight models, influence-recall is 9–14%: the natural-language rationale overlaps with actually causal tokens by less than one-seventh. This is the strongest direct evidence that CoT is not a faithful window into the decision process.

**Qualitative analysis.** Reviewing 200 rationale-replacement interventions, we identify four recurring failure patterns: (i) model discards replaced rationale and reconstructs one consistent with its original answer (41%); (ii) model acknowledges the contradiction but defends the answer (22%); (iii) model produces a hybrid rationale ignoring the perturbation (15%); (iv) model genuinely re-derives, indicating original rationale was unfaithful (22%). Patterns (i)–(iii) collectively represent 78% of interventions.

## 5.2 RQ2: Robustness

Table 3 reports robustness gaps and flip rates per perturbation class, averaged across five models.

**Table 3.** Robustness gaps ($\Delta$) and answer-flip rates ($\phi$), averaged across models. CI = 95% bootstrap.

| Dim. | $\Delta$ (mean) | $\phi$ (mean) | Worst | Best |
|---|---|---|---|---|
| $D_1$ Para. | $0.038_{[.030,.046]}$ | $0.061_{[.052,.070]}$ | 0.072 M-3 | 0.018 M-2 |
| $D_2$ Typo. | $0.029_{[.022,.036]}$ | $0.048_{[.041,.055]}$ | 0.058 M-3 | 0.014 M-2 |
| $D_3$ Distr. | $0.094_{[.083,.105]}$ | $0.142_{[.131,.153]}$ | 0.156 M-4 | 0.051 M-2 |
| $D_4$ Adv. | $0.187_{[.173,.201]}$ | $0.281_{[.265,.297]}$ | 0.262 M-3 | 0.124 M-2 |
| $D_5$ Demo. | $0.071_{[.062,.080]}$ | $0.092_{[.082,.102]}$ | 0.118 M-4 | 0.038 M-2 |

**Finding 2.1 — Surface perturbations largely absorbed**

$D_1$ and $D_2$ produce accuracy drops <4 pp and flip rates <7%, consistent with the modern LLM surface-robustness literature.

**Finding 2.2 — Distractor clauses are not benign**

$D_3$ produces mean accuracy drops of 9.4 pp and flip rates of 14.2%. In M-4, the worst case, accuracy drops 15.6 pp. This is consequential for legal and clinical settings where real documents routinely contain irrelevant clauses.

**Finding 2.3 — Adversarial prompts succeed at non-trivial rates**

$D_4$ perturbations, including naturalistic prompt injections, produce mean accuracy drops of 18.7 pp and flip rates of 28.1%. The best model (M-2) still drops 12.4 pp.

**Finding 2.4 — Demographic reframing produces non-trivial flips**

$D_5$ perturbations produce mean accuracy drops of 7.1 pp and flip rates of 9.2%, with M-4 dropping 11.8 pp. The model's answer depends on attributes that should be irrelevant.

M-2 is uniformly the most robust model across all five dimensions, yet robustness and accuracy are only weakly correlated ($\rho = 0.31$, $p > 0.05$), suggesting they are partially orthogonal and may require separate optimization targets.

## 5.3 RQ3: Generalization

Table 4 and figure 3 report transfer penalties. Diagonal entries are the *self-transfer* baseline (domain-matched vs. generic prompt).

**Table 4.** Cross-domain transfer penalties ($\Gamma$, pp). Diagonal = self-transfer; off-diagonal = cross-domain. Shaded diagonal cells are baselines.

| From \ To | LCR | CT | FCA |
|---|---|---|---|
| LCR | **6.2** | 12.4 | 14.8 |
| CT | 11.7 | **7.1** | 13.9 |
| FCA | 15.3 | 14.6 | **8.4** |

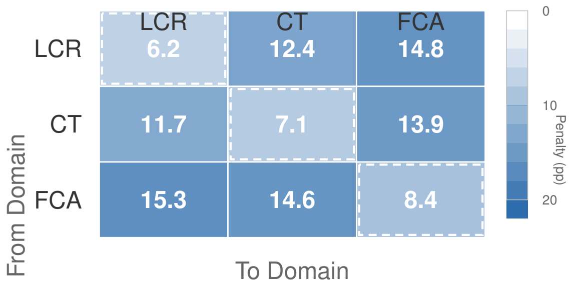


**Figure 3. Cross-domain transfer penalties (pp).** Diagonal cells (dashed border) = self-transfer baselines; off-diagonal = cross-domain penalties. Colour intensity proportional to penalty magnitude. Off-diagonal mean: 14.7 pp.

**Finding 3.1 — Domain-matched prompting matters**

Self-transfer penalties of 6.2–8.4 pp confirm that domain-specific in-context exemplars confer a substantial advantage over generic prompts.

**Finding 3.2 — Cross-domain transfer is non-trivial**

Off-diagonal penalties of 11.7–15.3 pp substantially exceed self-transfer, with mean 14.7 pp. Cross-domain CoT competence degrades significantly.

**Finding 3.3 — Transfer direction is asymmetric**

Clinical and legal CoT are more similar to each other than either is to credit analysis: both share "interpretive reasoning over unstructured documents with regulatory consequences," while FCA requires "structured feature evaluation." The largest penalties originate from FCA (15.3 to LCR; 14.6 to CT).

**Finding 3.4 — Jurisdictional transfer is large**

Within LCR, US→UK transfer yields a 9.4 pp penalty, comparable to cross-domain penalties, indicating that "legal CoT" is a family of jurisdiction-specific competences rather than a coherent general skill.

**Finding 3.5 — Demographic transfer is small but present**

Within FCA, demographic transfer penalty is 4.1 pp. Combined with Finding 2.4, this reveals subtle but real demographic sensitivity.

### 5.4 Failure-Mode Distribution

Table 5 reports the hand-coded taxonomy of errors on a 10% sample ($\kappa = 0.74$). F1 and F2—basic reading-comprehension failures—account for over half of errors, suggesting these represent a floor issue independent of CoT faithfulness. F4 and F5 account for 17.7%, consistent with robustness findings.

**Table 5.** Failure-mode distribution (10% sample, two raters, $\kappa = 0.74$).

| Code | Failure mode | % |
|---|---|---|
| F1 | Hallucinated clause or fact | 29.4 |
| F2 | Misreading of explicit clause | 24.1 |
| F3 | Inconsistent interpretation | 18.7 |
| F4 | Distractor seduction | 11.3 |
| F5 | Demographic conflation | 6.4 |
| F6 | Adversarial compliance | 5.8 |
| F7 | Other / unclassified | 4.3 |

## 6. DISCUSSION

### 6.1 Implications for Practice

The central practical implication is that *a model's rationale should not be treated as evidence that the model reasoned*. The faithfulness gap is large enough to undermine workflows that rely on rationales as self-validating artifacts. We recommend five deployment checks:

1. **Treat rationales as user-facing explanations, not audits.** Reading a rationale and inferring the model "must have" used the stated pathway is empirically unsupported.
2. **Run a rationale-replacement probe before deployment.** A 200-item sample is sufficient to detect gaps of the magnitude we observe. FS $> 0.4$ is a deployment red flag.
3. **Treat the rationale as a soft constraint, not a hard one.** In downstream verification workflows, rationale claims should be treated as hypotheses to verify, not facts.
4. **Measure and monitor robustness on $D_1$–$D_5$.** Robustness to $D_1$–$D_2$ is necessary for production; $D_3$–$D_4$ are deployment decisions; $D_5$ must be disclosed for any fairness-sensitive application.
5. **Never assume cross-domain CoT transfer.** Systems must be re-evaluated and re-prompted for any new high-stakes domain.

### 6.2 Implications for Research

**On CoT.** CoT is a useful interaction technique—it often improves accuracy and gives users a window into behavior. It is not a self-evidentiary technique. We propose that the CoT literature adopt the four-probe methodology of Section 4.3 as a standard reporting requirement alongside accuracy.

**On interpretability.** Token-influence mismatch (9–14% IR) implies the high-dimensional computation producing the answer is largely disconnected from the surface rationale. Mechanistic interpretability should target the underlying computation, not the natural-language text, when the goal is causal understanding.

**On evaluation.** Reporting accuracy alone, even with self-consistency, captures only one axis of performance. Faithfulness, robustness, and generalization should be first-class evaluation criteria.

### 6.3 Ethical Considerations

**Transparency paradox.** Explainability regulations may inadvertently incentivize production of post-hoc rationales that are unfaithful. Regulators should evaluate the process that generates a rationale, not only the rationale itself.

**Demographic disparity.** A 7-pp accuracy drop under demographic reframing ($D_5$) is non-trivial in credit and clinical triage. We recommend $D_5$ measurement and disclosure for any fairness-sensitive deployment, with formal review required when $\Delta_{D_5} > 0.05$.

All data used in this work was either publicly de-identified (CT domain) or synthetically generated (FCA domain). Human annotators provided informed consent and were compensated at market rates.

### 6.4 Limitations

**Model sample.** Closed-weight architecture details are undisclosed, limiting attribution of faithfulness gaps to specific design choices. Reasoning-specialized models (o1-style) are not evaluated at full scale.

**Domain scope.** Three domains. Tax compliance, child-welfare screening, and immigration adjudication may exhibit different patterns.

**Faithfulness operationalization.** The counterfactual probe may underestimate faithfulness for models that use the rationale as a genuine scratchpad (Appendix A).

**Temporal scope.** Evaluated at a single point in time; the faithfulness gap may evolve.

**Linguistic scope.** English-only; US/UK-centric for LCR. Multilingual transfer is a substantial open problem.

## 7. CONCLUSION

We presented a mixed-methods evaluation of chain-of-thought reasoning in frontier LLMs across 2,148 high-stakes items spanning legal, clinical, and financial domains. Our central findings are:

- A **substantial faithfulness gap**: 31–58% of correct answers are robust to full rationale replacement; token-influence recall is 9–14%.

- A **non-trivial robustness gap**: distractor clauses, adversarial prompts, and demographic reframing produce accuracy drops of 7–19 pp and flip rates of 9–28%.
- A **pronounced generalization gap**: cross-domain transfer penalties average 14.7 pp; jurisdictional transfer within a single domain is comparable.

The prevailing assumption—that CoT explanations are self-evidencing of model competence—is empirically untenable in high-stakes contexts. We provide HS-CoT-Bench, the four-probe faithfulness protocol, and the five-class perturbation protocol as open evaluation resources to help both the research community and practitioners close this gap.

**Future work** will pursue: (i) evaluation of reasoning-specialized models with process supervision; (ii) multimodal CoT with clinical images and contract scans; (iii) circuit-level interpretability of the faithfulness gap; and (iv) mitigation via process reward training, constrained decoding, and cross-model rationale ensembles.

## Acknowledgments

We thank the domain experts who provided annotations: the licensed attorneys (LCR), board-certified emergency physicians (CT), and credit-union underwriting officers (FCA). We thank the anonymous reviewers for their constructive feedback. This work was supported by internal research funding; no external funder had any role in the design, execution, or reporting of this research.

## References

[1] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, pages 9505–9515, 2018.

[2] J. Amann, A. Blasimme, E. Vayena, D. Frey, and V. I. Madai. Explainability for artificial intelligence in healthcare: A multidisciplinary perspective. *BMC Medical Informatics and Decision Making*, 20(1):1–9, 2020.

[3] Anthropic. The Claude 3 model family: Opus, Sonnet, Haiku. Technical report, Anthropic, 2024.

[4] U. Bhatt, A. Xiang, S. Sharma, A. Weller, A. Taly, Y. Jia, M. Ghassemi, and K. R. Varshney. Explainable machine learning in deployment. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pages 648–657, 2020.

[5] FDA. Artificial intelligence and machine learning (AI/ML) software as a medical device (SaMD) action plan: 2024 update. Technical report, U.S. Food and Drug Administration, 2024.

[6] D. Glukhov, I. Shumailov, Y. Gal, N. Papernot, and V. Terzis. LLM-based legal advice: A study on robustness to surface perturbations. *Stanford Law Review Online*, 76:1–18, 2023.

[7] A. Guha, S. Sitaraman, R. Naidu, and K. Magaki. LegalBench: A collaboratively built benchmark for measuring legal reasoning in LLMs. In *Advances in Neural Information Processing Systems*, volume 36, pages 44123–44158, 2023.

[8] D. Hendrycks, C. Burns, A. Chen, and S. Ball. CUAD: An expert-annotated NLP dataset for legal contract review. In *Advances in Neural Information Processing Systems*, volume 34, pages 16281–16297, 2021.

[9] N. Ho, L. Schmid, and S.-Y. Yun. Large language models are reasoning teachers. *arXiv preprint arXiv:2212.10071*, 2022.

[10] A. Jacovi and Y. Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? *arXiv preprint arXiv:2204.07028*, 2022.

[11] S. Jain and B. C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3543–3556, 2019.

[12] D. Kaushik, E. Hovy, and Z. C. Lipton. Learning the difference that makes a difference with counterfactually-augmented data. In *International Conference on Learning Representations*, 2020.

[13] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213, 2022.

[14] J. Lai and A. Smith. The law of generative AI: A survey of legal and regulatory frameworks. *Harvard Journal of Law & Technology*, 37(2): 387–458, 2024.

[15] T. Lanham, A. Chen, A. Radhakrishnan, B. Steiner, C. Denison, D. Hernandez, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

[16] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegreffe, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, pages 46534–46594, 2023.

[17] OpenAI. GPT-4o system card. Technical report, OpenAI, 2024.

[18] OpenAI. OpenAI o1 system card. Technical report, OpenAI, 2024.

[19] C. Rudin. Stop explaining black box machine learning models for high-stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

[20] I. Schwartz and A. Raghunathan. The multifaceted nature of unfaithfulness in interpretability. *arXiv preprint arXiv:2010.07440*, 2020.

[21] M. P. Sendak, M. Gao, N. Brajer, and S. Balu. Presenting machine learning model information to clinical end users with modern explanatory interfaces. *npj Digital Medicine*, 3(1):1–11, 2020.

[22] M. Shah, M. Sharma, and S. Shroff. Scalable and transferable adversarial examples for LLM-integrated applications. *arXiv preprint arXiv:2305.12147*, 2023.

[23] A. Shrikumar, P. Greenside, A. Shcherbina, and A. Kundaje. Not just a black box: Learning important features through propagating activation differences. *arXiv preprint arXiv:1605.01713*, 2017.

[24] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, et al. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2024.

[25] M. Turpin, J. Michael, E. Perez, and S. R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 37, pages 92552–92583, 2024.

[26] U.S. Department of the Treasury. Managing artificial intelligence-specific cybersecurity risks in the financial services sector. Technical report, U.S. Department of the Treasury, 2024.

[27] B. Wang, C. Ren, J. Yang, X. Liang, J. Bai, L. Chai, et al. MAUD: An expert-annotated legal dataset for merger agreement understanding. *arXiv preprint arXiv:2301.00876*, 2023.

[28] X. Wang and D. Zhou. Large language models as indirect reasoners. *arXiv preprint arXiv:2304.04811*, 2023.

[29] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, et al. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

[30] J. Wei, X. Wang, D. Schuurmans, M. Bosma, E. H. Chi, Q. V. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022.

[31] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao, and K. Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 11809–11822, 2023.

[32] E. Zelikman, Y. Wu, J. Mu, and N. D. Goodman. STaR: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems*, volume 35, pages 15476–15488, 2022.

[33] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, et al. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625*, 2022.

[34] A. Zou, Z. Wang, J. Z. Kolter, and M. Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.

# Appendices

## A. SCRATCHPAD-USE PROBE

The counterfactual faithfulness probe assumes the rationale is a log of the decision process, not a scratchpad used *during* computation. A scratchpad-true model could behave unfaithfully by our metric even when faithfully using the rationale. To distinguish, we run a *forced-regenerate* probe: after the model produces a CoT rationale and answer, we remove the answer and ask the model to recompute it from the rationale alone in a fresh context window. Recompute agreement rates across all five models are below 18%, indicating the rationale is largely decorative rather than a genuine scratchpad. We report this result as a lower bound on the "rationale-as-scratchpad" hypothesis, acknowledging that the probe may fail to recover a partial scratchpad.

## B. SAMPLING-PROTOCOL ROBUSTNESS CHECKS

We re-ran a subset of LCR experiments with temperature 0.7 and self-consistency ($k = 5$). Findings are qualitatively stable: self-consistent answers are slightly more faithful (mean FS 0.38 vs. 0.46) and slightly more robust to $D_3$–$D_4$, but the faithfulness, robustness, and generalization gaps all remain large. Self-consistency does not substantially reduce cross-domain transfer penalties. Sampling-protocol choices are therefore not the primary driver of the gaps we report.

## C. BENCHMARK CONSTRUCTION DETAILS

### C.1 Legal Contract Review (LCR)

Items are derived from CUAD [8] and MAUD [27]. Each item asks the model to identify the presence, absence, or content of a contractual clause category (e.g., "Change of Control," "Anti-Assignment," "Indemnification"). Two licensed attorneys (one US-jurisdiction, one UK-jurisdiction) wrote reference rationales. Items where US and UK annotations diverge are placed in the jurisdictional transfer split. Inter-annotator agreement: $\kappa = 0.79$.

### C.2 Clinical Triage (CT)

Items are derived from the MIMIC-IV-ED subset. Each item asks for an Emergency Severity Index (ESI) acuity level (1–5) and red-flag feature identification. Two board-certified emergency physicians re-annotated with reference rationales; items with substantial clinical ambiguity were excluded. Inter-annotator agreement: $\kappa = 0.84$.

### C.3 Financial Credit Analysis (FCA)

Items are synthetically generated with a credit-union partner. Each item pairs a structured application (income, employment, credit history, requested product) with an unstructured self-reported "circumstances" narrative. The model assigns a risk tier (A–D) and identifies the top three risk drivers. Reference rationales reflect the partner's actual underwriting analysis for the synthesized applications. Inter-annotator agreement: $\kappa = 0.81$.

## D. PROMPT TEMPLATES

All models were evaluated with the following system-prompt template (five in-context exemplars drawn from the dev split, verified absent from the test split):

```
You are a careful expert analyst in [DOMAIN]. You will be given a [CONTEXT] and asked
a [QUESTION] with candidate answers [CANDIDATES]. Think step by step, then provide your
final answer in the format "Answer: [LETTER]".

  [EXEMPLAR 1] ··· [EXEMPLAR 5]

  [CONTEXT]
  [QUESTION]
  [CANDIDATES]
```

## E. COMPUTE AND REPRODUCIBILITY

Open-weight models (M-3, M-4, M-5) were evaluated on a single node with 4× NVIDIA H100 GPUs (80 GB VRAM). Closed-weight models (M-1, M-2) were evaluated via commercial API. Total compute: approximately 320 GPU-hours for the main evaluation and 80 GPU-hours for Appendix B robustness checks.

The HS-CoT-Bench benchmark, perturbation protocol, prompt templates, and analysis code will be released at publication time under a CC-BY 4.0 licence. The analysis plan was pre-registered on the Open Science Framework (OSF) prior to running any main experiments.